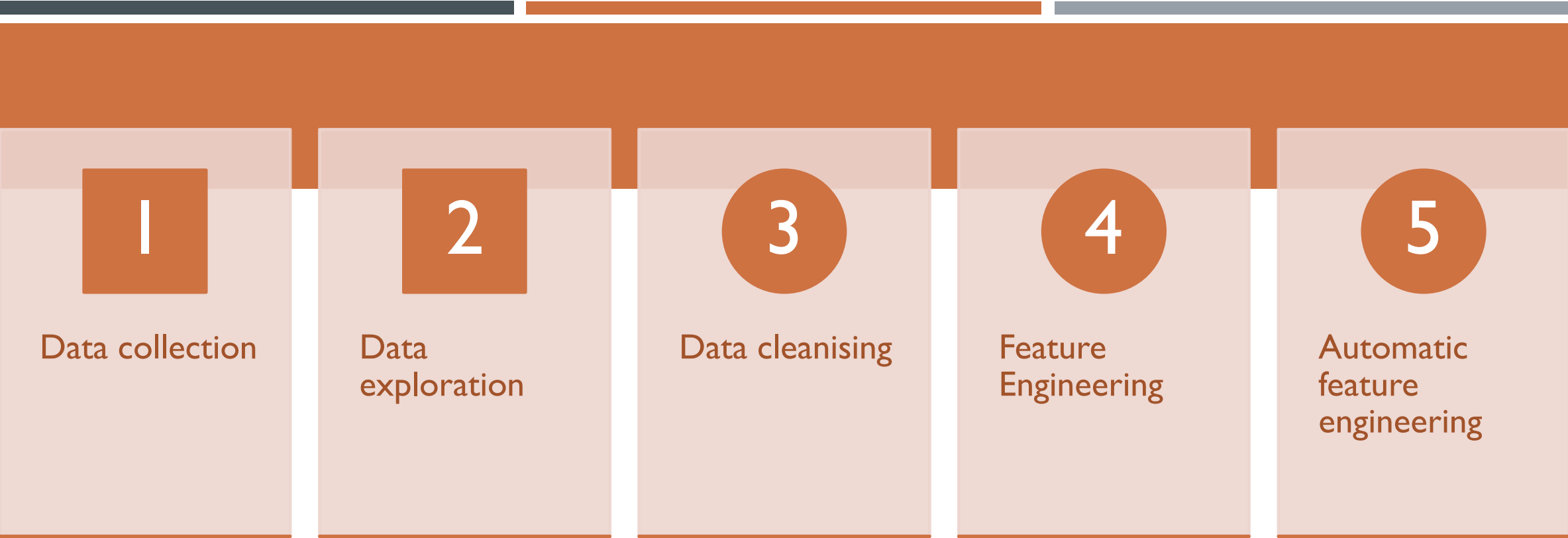




DATA PREPARATION



1

Data collection

2

Data
exploration

3

Data cleansing

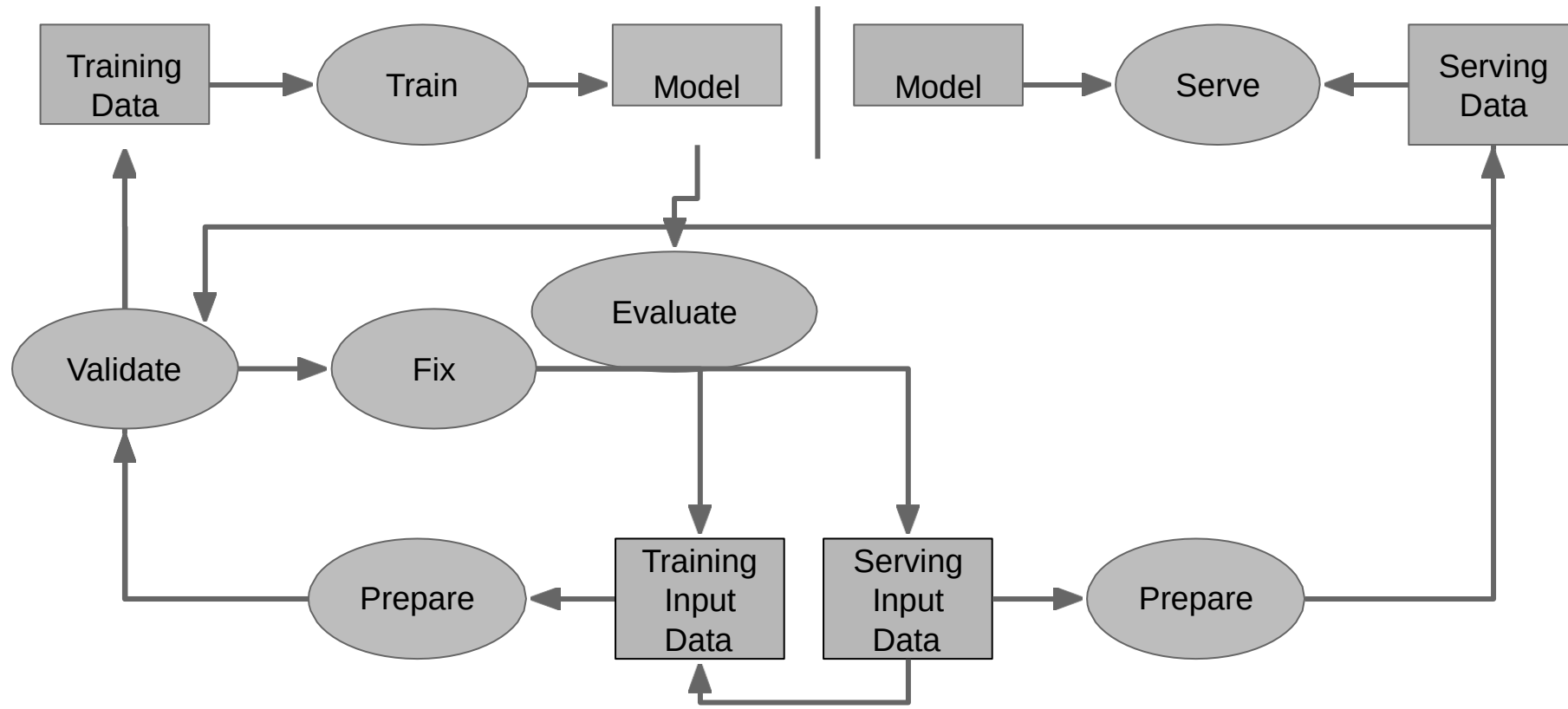
4

Feature
Engineering

5

Automatic
feature
engineering

DATA UNDERSTANDING IN ML PIPELINE



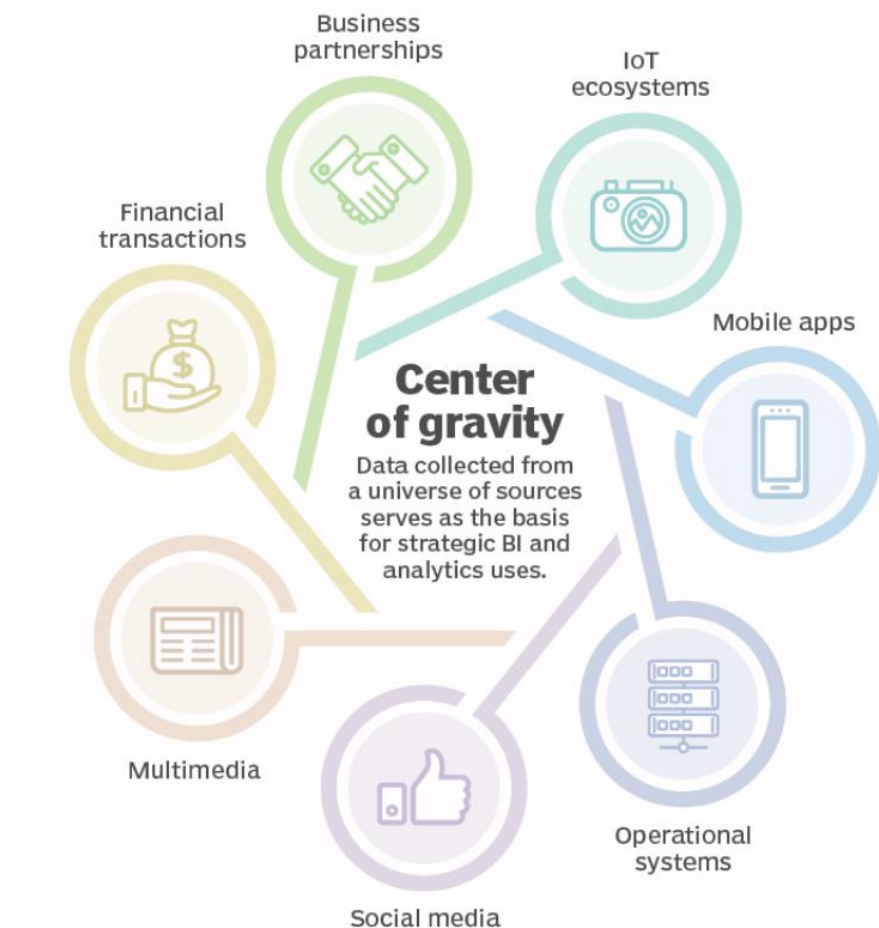


DATA COLLECTION



DATA COLLECTION

- Data collection is the process of gathering data for use in ML pipeline
- Operational databases regularly collect data on customers, employees, sales and other aspects of business operations when transactions are processed and data is entered.
 - Data as “side effect” of an administrative or business act
- It also possible to conduct surveys and track social media to get feedback from customers.
- Type of data source
 - Primary data source
 - Secondary data source



PRIMARY DATA SOURCE

- Primary data is a type of data that is collected directly from main sources through accessing data, interviews, surveys, experiments, etc.
- Primary data are usually collected from the source—where the data originally originates from and are regarded as the best kind of data.
- The sources of primary data are usually chosen and tailored specifically to meet the demands or requirements of particular research.
- Primary data source can be very expensive
 - For example, when doing a market survey, the goal of the survey and the sample population need to be identified first.
 - an offline survey will be more suitable for a population living in remote areas without an internet connection compared to online surveys

SECONDARY DATA

- Secondary data is data that has been collected in the past by someone else but made available for others to use.
- They are usually once primary data but become secondary when used by a third party.
- Secondary data are usually
 - easily accessible because they are mostly shared publicly.
 - not tailored specifically to meet your needs.
- Some common sources of secondary data include open data, government statistics, integrated data from data warehouse
 - In most cases, these sources cannot be trusted as authentic.

BIAS, FAIRNESS

- Machine learning bias is the tendency of the algorithm to incorporate and reflect human biases.
- ML models and their predictions are heavily determined by the data used for training
- if the data is not high-quality and scrubbed, it can include biases.
- Skewed datasets comprise incomplete, inaccurate, and inappropriate training data that produce poor decisions.
- Historical bias
 - ML for loan applicants could show preferential treatment toward men versus women.
- Representation bias
 - E.g. facial recognition for Asian persons
- Measurement, evaluation, aggregation bias

DATA COLLECTION

- Key questions to ask when looking for training data include:
 1. How will the resulting ML model be used? (On what population will the model be making predictions on and when)
 2. What are hypothetical edge cases or high-stakes scenarios where we really need model make the right prediction?

ERRORS IN PREDICTION

- Beery et al. [\[BGP18\]](#).
 - The left-hand image is correctly predicted to contain a cow by a trained image classifier,
 - this model fails to produce the same prediction for the right-hand image.
- cows were always pictured in grass fields in the dataset used to train the classifier

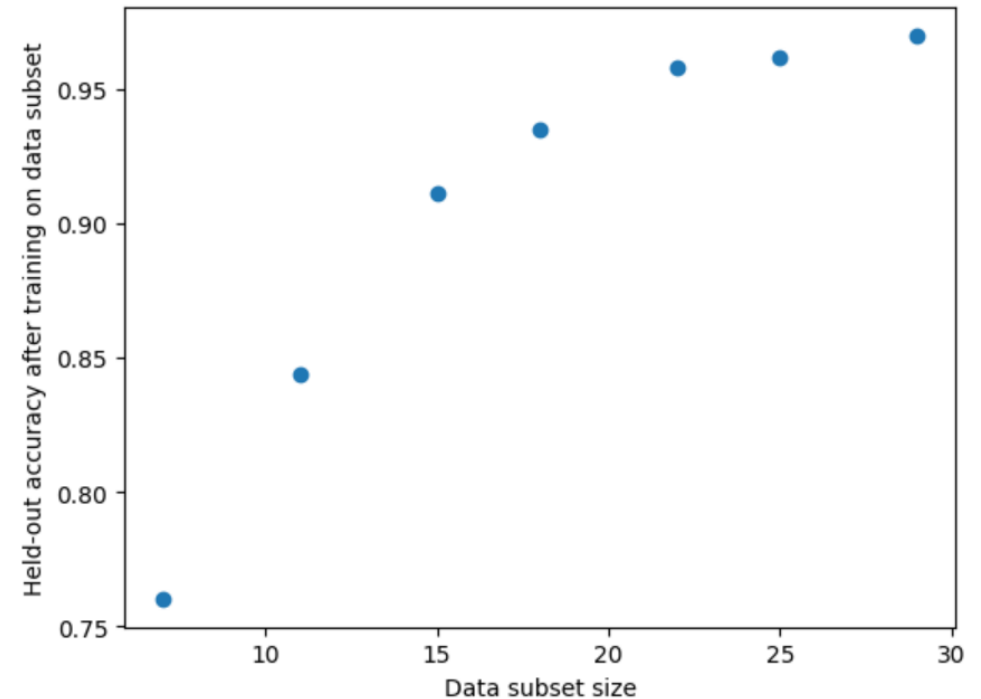


SELECTION BIAS

- systematic deviation between the distribution from which our collected training data stem and the actual distribution where the model will be deployed
 - Dataset fails to be representative of the real-world setting
- Time/location bias. training data collected in the past for applications where future data will look different, or collected from one country for a model deployed world-wide
 - Demographics bias. training data about people in which a particular minority group is under-represented
 - Response bias. Survey response rates
 - Availability bias- Using data that happens to be convenient rather than most representative
 - only surveying interviewees that are my friends.
 - Long tail bias. in applications with many different possible rare scenarios

BIGGER IS BETTER

- The task of predicting accuracy at n' requires extrapolation,
- An empirical observation [\[RRB20\]](#) that the performance of a model is remarkably predictable according to:
$$\log(\text{error}) = -a \cdot \log(n) + b$$
- We can estimate the scalar parameters a, b by minimizing the mean squared error over observations $\{(n_j, 1 - a_{ij})\}$
- Subsequently just plug in n' to the above formula to predict the error expected for a model trained on a larger dataset of this size.



DATA PROVENANCE

- It is mandatory to know the provenance of data are you using
- Data provenance refers to a record trail that
 - accounts for the origin of a piece of data (in a database, document or repository)
 - together with an explanation of how and why it got to the present place.

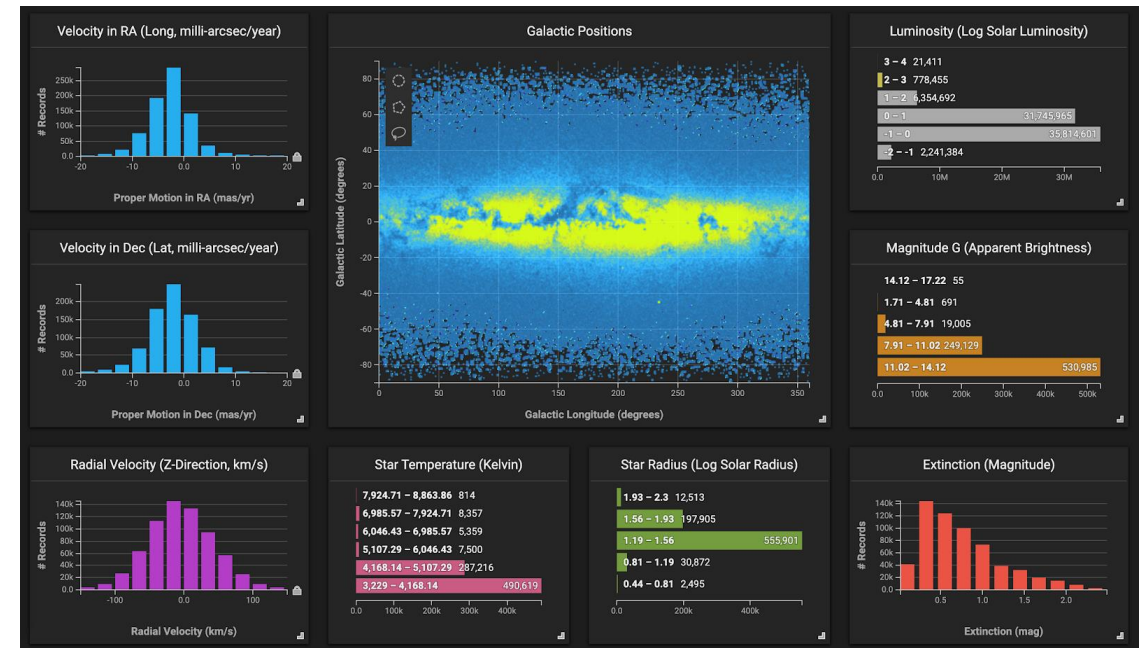


DATA EXPLORATION



DATA EXPLORATION

- Data exploration refers to the initial step in data analysis
- Data exploration techniques include
 - manual analysis and automated data exploration software solutions
- To visually explore and identify
 - relationships between different data variables,
 - the structure of the dataset,
 - the presence of outliers,
 - the distribution of data values



DATA EXPLORATION

- Tabular data may show different types of data for different columns (or variables or features)
- Continuous
 - The price of houses, temperature, speed
- Discrete
 - age, points in a games
- Categorical data
 - types of data which may be divided into groups.
 - race, sex,
 - age group, and educational level may also be considered in a numerical manner by grouping age and highest grade completed,
- Count data
 - the number of children a couple has, the number of doctors visits per year a person makes, and the number of trips per month that a person takes.
- Both categorical, and count data are «discrete» from a data type view point but they have different meanings

DATA EXPLORATION

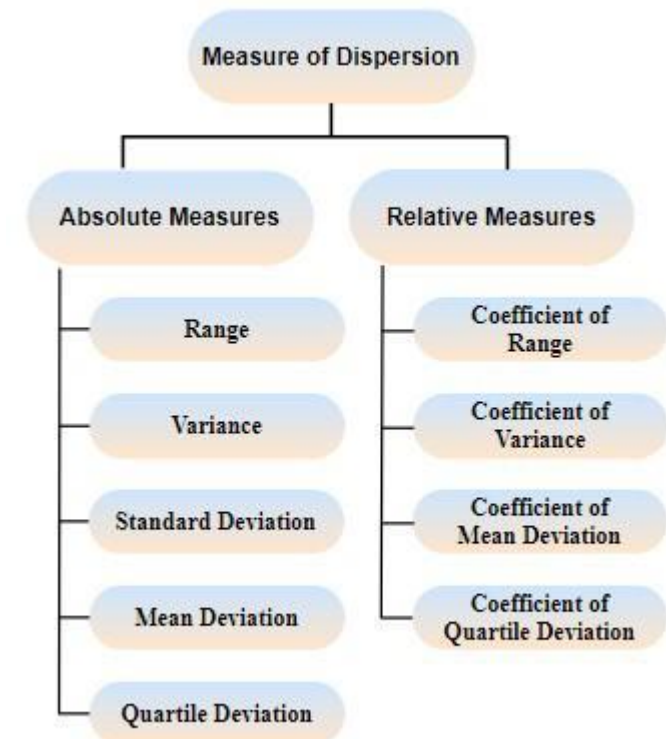
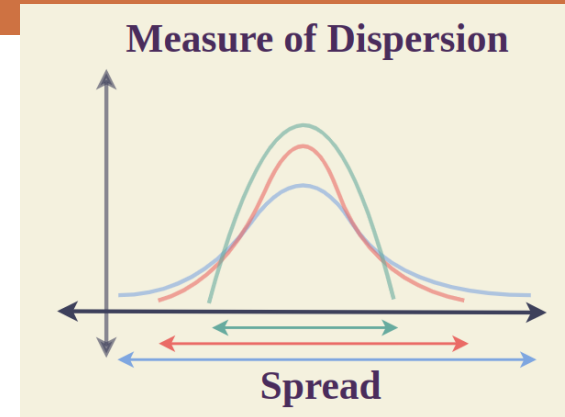
- Data exploration steps to follow before building a machine learning model include:
- Variable identification: define each variable and its role in the dataset
- Univariate analysis:
 - continuous and discrete variables, build box plots or histograms for each variable independently;
 - categorical and count variables, build bar charts to show the frequencies
- Bi-variable analysis - determine the interaction between variables by building visualization tools
 - Continuous and Continuous: scatter plots
 - Categorical and Categorical: stacked column chart
 - Categorical and Continuous: boxplots combined with swarmplots
- Detect and treat missing values
- Detect and treat outliers

EXPLORATIVE DATA ANALYSIS

- Exploratory Data Analysis (EDA) is a statistical technique to analyze data sets for their broad characteristics
- Dispersion measure
- Box plot
- Q-Q-plot
- Bivariate scatter plot
- swarmplots

MEASURE OF DISPERSION

- Measures of Dispersion measure the scattering of the data
- How the values are distributed in the data set.
- Typical measure of dispersion are
 - Standard Deviation
 - Mean Deviation
 - Quartile Deviation
 - Variance
 - Range, etc



BOX PLOT

- A box and whisker plot (box plot)
- displays the five-number summary of a set of data.
 - the minimum, first quartile, median, third quartile, and maximum.
- In a box plot, we draw a box from the first quartile to the third quartile.
- A vertical line goes through the box at the median.
- The whiskers go from each quartile to the minimum or maximum.

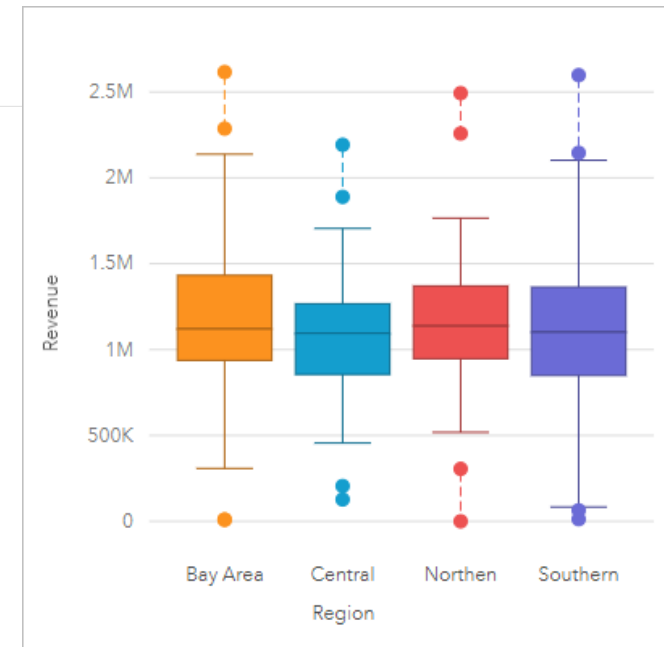
```
# Import libraries
import matplotlib.pyplot as plt
import numpy as np

# Creating dataset
np.random.seed(10)
data = np.random.normal(100, 20, 200)

fig = plt.figure(figsize =(10, 7))

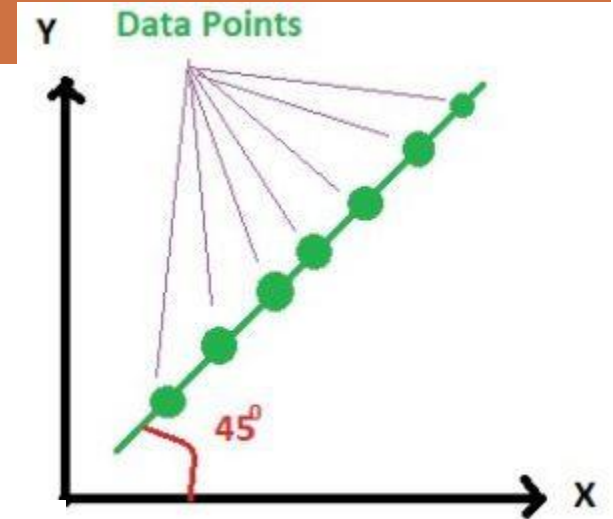
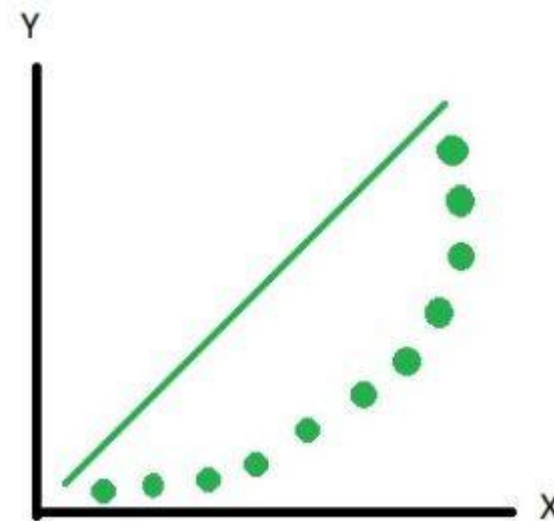
# Creating plot
plt.boxplot(data)

# show plot
plt.show()
```



QQ PLOT

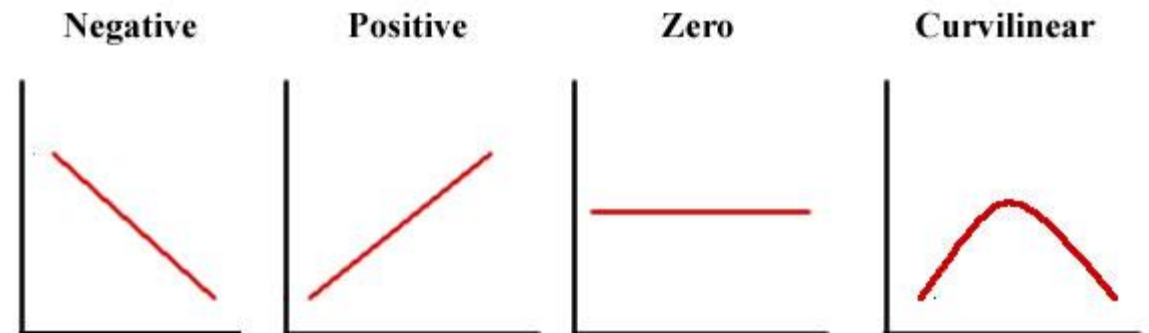
- The quantile-quantile plot is a graphical tool to help us assess if a set of data plausibly came from some theoretical distribution such as a normal or exponential
- This plot provides a summary of whether the distributions of two variables are similar or not with respect to the locations.



SCATTER PLOT

- A scatter plot is the graphic representation of the relationship between the two variables coming from a bivariate data set
- It permits you to see at a glance the degree and pattern of relation between the two variables.
- It is used to evaluate if a feature can be a good predictor for a target variable

```
import pandas as pd  
  
reviews = pd.read_csv("winemag-data_first150k.csv", index_col=0)  
reviews[reviews['price'] < 100].sample(100).plot.scatter(x='price', y='points')
```



SWARMPLOT

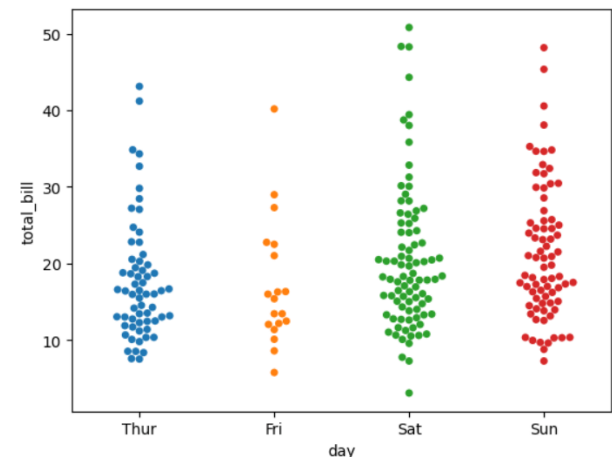
- swarmplot is a type of categorical scatter plot used to visualize the distribution of data points in a dataset. It is a useful tool for exploring and analyzing the relationship between two or more variables in a dataset
- In a swarmplot, each data point is represented as a dot, with the dots arranged in such a way that they do not overlap with each other.
- This arrangement helps to avoid the problem of overplotting, which can occur when multiple data points are plotted on top of each other, making it difficult to distinguish between them.

```
import seaborn as sns
```

```
# set color palette sns.set_palette("husl")
```

```
■ # create swarmplot
```

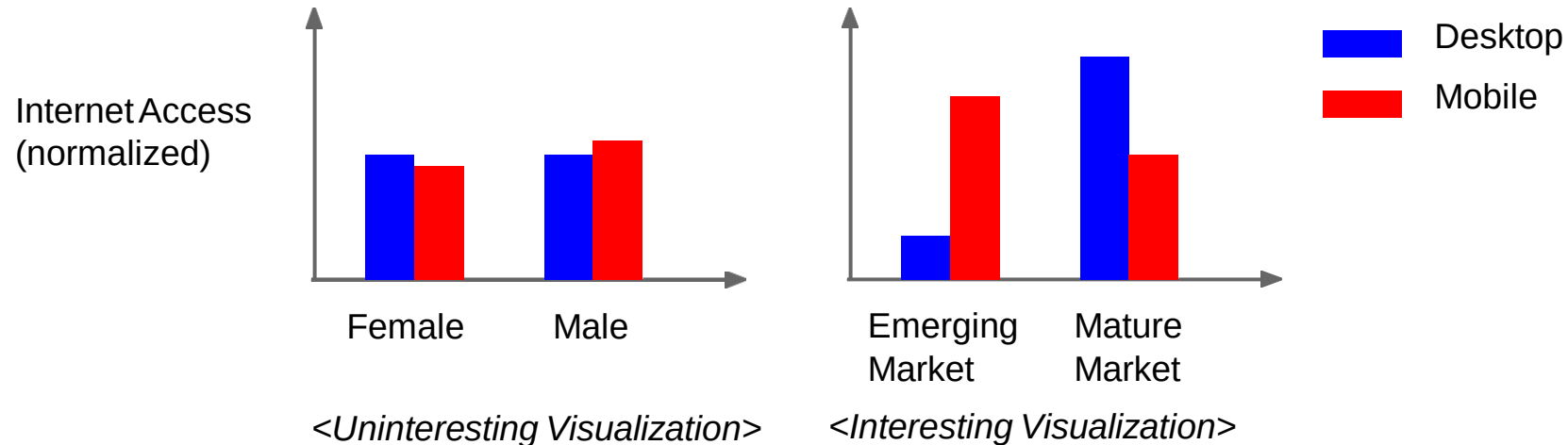
```
■ sns.swarmplot(x="category", y="value", data=data)
```



Type of Variables (Vs.)	Categorical (incl. discrete numerical)	Continuous
Categorical (incl. discrete numerical)	- Frequency of the two categories/ other continuous variables' range - Crosstab - Heatmaps - Stacked bar charts	- Range of continuous variable with respect to each category - Boxplots - Violin plots - Swam plots - Count plots - Bar plot
Continuous	- Range of continuous variable with respect to each category - Boxplots - Violin plots - Swam plots - Count plots - Bar plot	- How the increase or decrease in one variables changes with the other - Scatterplot - Line plots

DATA-DRIVEN VISUALIZATION

- Recommends “interesting” visualizations using a deviation-based metric
- Research question: what is the confidence of these visualizations?



- Do you know how to create interesting visualizations?
 - <https://microsoft.github.io/lida/>

TOOLS

EDA



Interactive GUI-based tools

Tableau

PowerBI

SPSS

JMP



Python open-source libraries

SweetViz

Lux

Dtale

Dabl

Bamboolib

PandasGUI

Facets



Data-Centric AI Visualization

VizML

DeepEye

Data2Vis

ExplainedED

FEDEX



DATA CLEANSING



HOW TO FILL MISSING VALUES?

- Deleting Rows with missing values
 - Missing values can be handled by deleting the rows or columns having null values. If columns have more than half of the rows as null then the entire column can be dropped.
- Impute missing values for continuous variable
 - Columns in the dataset which are having numeric continuous values can be replaced with the mean, median, or mode of remaining values in the column
- Impute missing values for categorical variable
 - the missing values can be replaced with the most frequent category. If the number of missing values is very large then it can be replaced with a new category.
- Other Imputation Methods
 - longitudinal behavior, it might make sense to use the last valid observation to fill the missing value. This is known as the Last observation carried forward (LOCF) method.
 - For the time-series dataset variable, it makes sense to use the interpolation of the variable before and after a timestamp for a missing value.
- Using ML Algorithms that support missing values

OUTLIERS

- Outliers are data points that differ significantly from the majority of the data points in a dataset
- Outliers can occur for various reasons, such as
 - measurement errors, data entry errors, sensor malfunctions, or
 - genuinely unusual observations in the underlying data generating process.
- Outliers can distort statistical measures, such as mean and standard deviation, leading to inaccurate summaries of the data and misleading conclusions.
- Detect outliers
- Handling Techniques:
 - **Imputation:** Imputation involves replacing or filling in the outlier values with estimated or imputed values.
 - **Transformation:** Transformation involves applying mathematical transformations to the data to reduce the impact of outliers. For example, taking the logarithm,
 - **Winsorizing:** Winsorizing involves capping or truncating the extreme values at a certain threshold.
 - **Data partitioning:** Another approach is to partition the data into different subsets based on the presence of outliers.



FEATURE ENGINEERING



FEATURE ENGINEERING

- Feature engineering is the process of transforming raw data into features that can better represent an underlying problem for computational predictive models, resulting in improved model accuracy on unseen data.
- These features can be directly fed to ML models to train them and make predictions.

FEATURE ENGINEERING

- Powerful feature transformations (like target encoding) can introduce leakage when applied wrong
- Usually requires domain knowledge about how features interact with each other
- Time-consuming, need to run thousand of experiments
- Why Feature Engineering matters
 - Extract more new gold features, remove irrelevant or noisy features
 - Simpler models with better results

FEATURE ENCODING

- Turn categorical features into numeric features to provide more fine-grained information
- Help explicitly capture non-linear relationships and interactions between the values of features
- Most of machine learning tools only accept numbers as their input
e.g., xgboost, gbm, glmnet, libsvm, liblinear, etc.

DATA TRANSFORMATION

- Raw data is often not ready for model training due to potential issues such as
 - noise, inconsistencies, and unnecessary information, leading to inaccurate
 - and biased results.
- **Mandatory transformations** for data compatibility.
 - Converting non-numeric features into numeric.
 - Resizing inputs to a fixed size.
- **Optional quality transformations** that may help the model perform better.
 - Tokenization or lower-casing of text features.
 - Normalized numeric features.
 - Allowing linear models to introduce non-linearities into the feature space.

WHEN TRANSFORM

- **Transforming prior to training**

- **Pros**

- Computation is performed only once.
 - Computation can look at entire dataset to determine the transformation.

- **Cons**

- Transformations need to be reproduced at prediction time. Beware of skew!
 - Any transformation changes require rerunning data generation, leading to slower iterations.

- **Transforming within the model**

- **Pros**

- Easy iterations. If you change the transformations, you can still use the same data files.
 - You're guaranteed the same transformations at training and prediction time.

- **Cons**

- Expensive transforms can increase model latency.
 - Transformations are per batch.

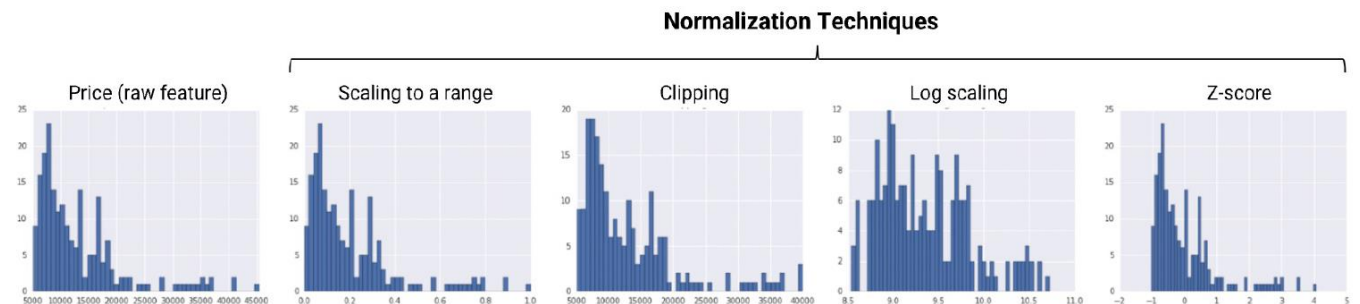
TRANSFORMATION TECHNIQUES

Transforming continuous data

- **Normalizing** - transforming numeric data to the same scale as other numeric data.
 - scaling to a range
 - clipping
 - log scaling
 - z-score is a variation of scaling that represents the number of standard deviations away from the mean.
- **Bucketing** - transforming numeric (usually continuous) data to categorical data

Clipping

- If data set contains extreme outliers.
- Limits or caps the values of a feature so that they do not exceed a certain predefined threshold, either on the high or low end.
- This threshold is often set based on domain knowledge or the characteristics of the data

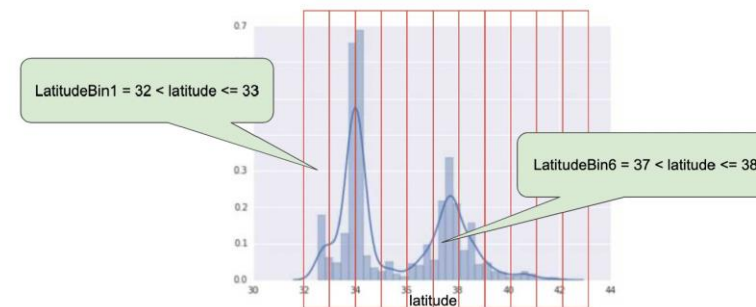
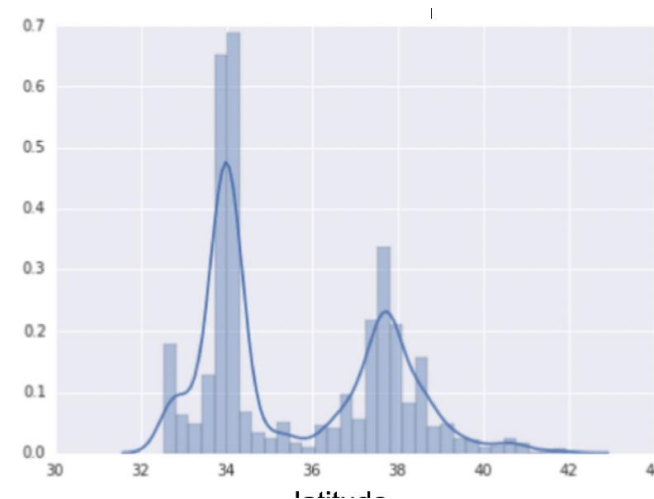


NORMALIZATION TECHNIQUES

Normalization Technique	Formula	When to Use
Linear Scaling	$x' = (x - x_{min}) / (x_{max} - x_{min})$	When the feature is more-or-less uniformly distributed across a fixed range.
Clipping	if $x > \max$, then $x' = \max$. if $x < \min$, then $x' = \min$	When the feature contains some extreme outliers.
Log Scaling	$x' = \log(x)$	When the feature conforms to the power law.
Z-score	$x' = (x - \mu) / \sigma$	When the feature distribution does not contain extreme outliers.

BUCKETING

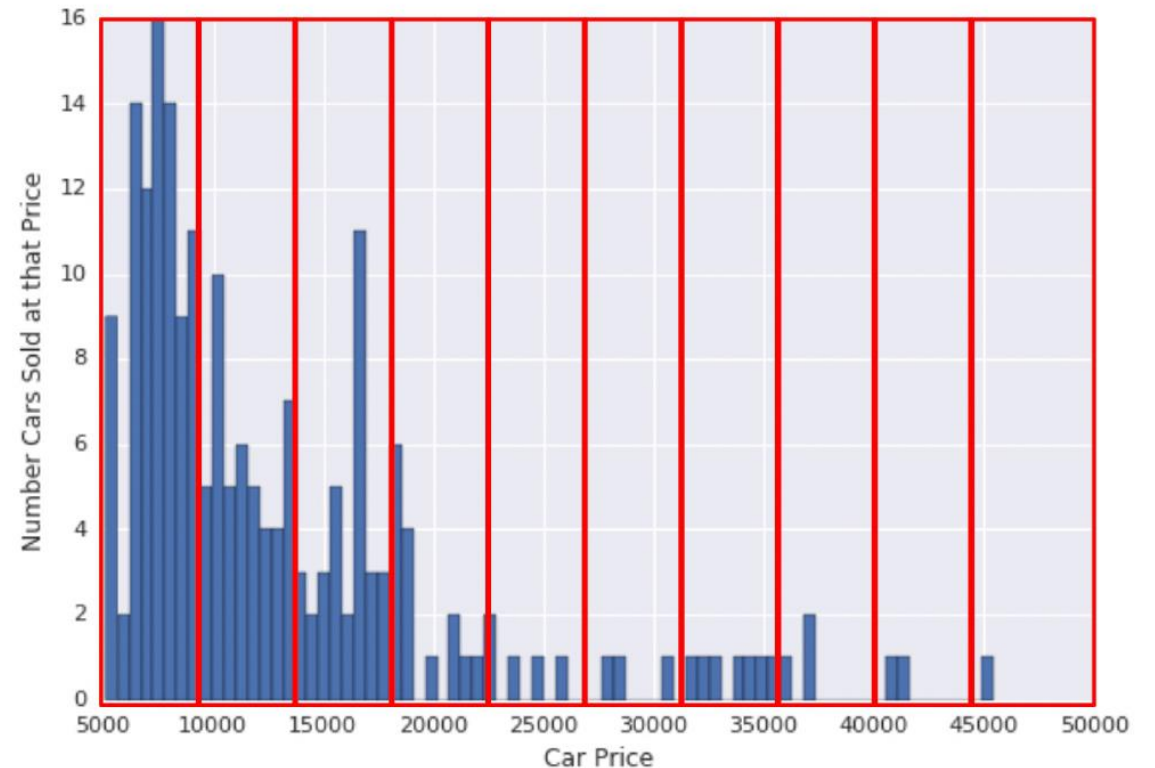
- Converting a single feature into multiple binary features called buckets or bins.
- Temperature values are divided into cold, warm and hot buckets based on their value. The values of each bucket are treated the same.



In this bucketing example, the boundaries are equally spaced.

QUANTILE BUCKETING

- For the car price dataset , the idea is to group car prices into specific ranges or "buckets" to make it easier to analyze and model the data.
- If these buckets are created with equally spaced intervals (e.g., \$5,000 intervals), it might not capture the underlying distribution of car prices effectively.
- This is especially true when the dataset has a highly skewed distribution with a wide range of values

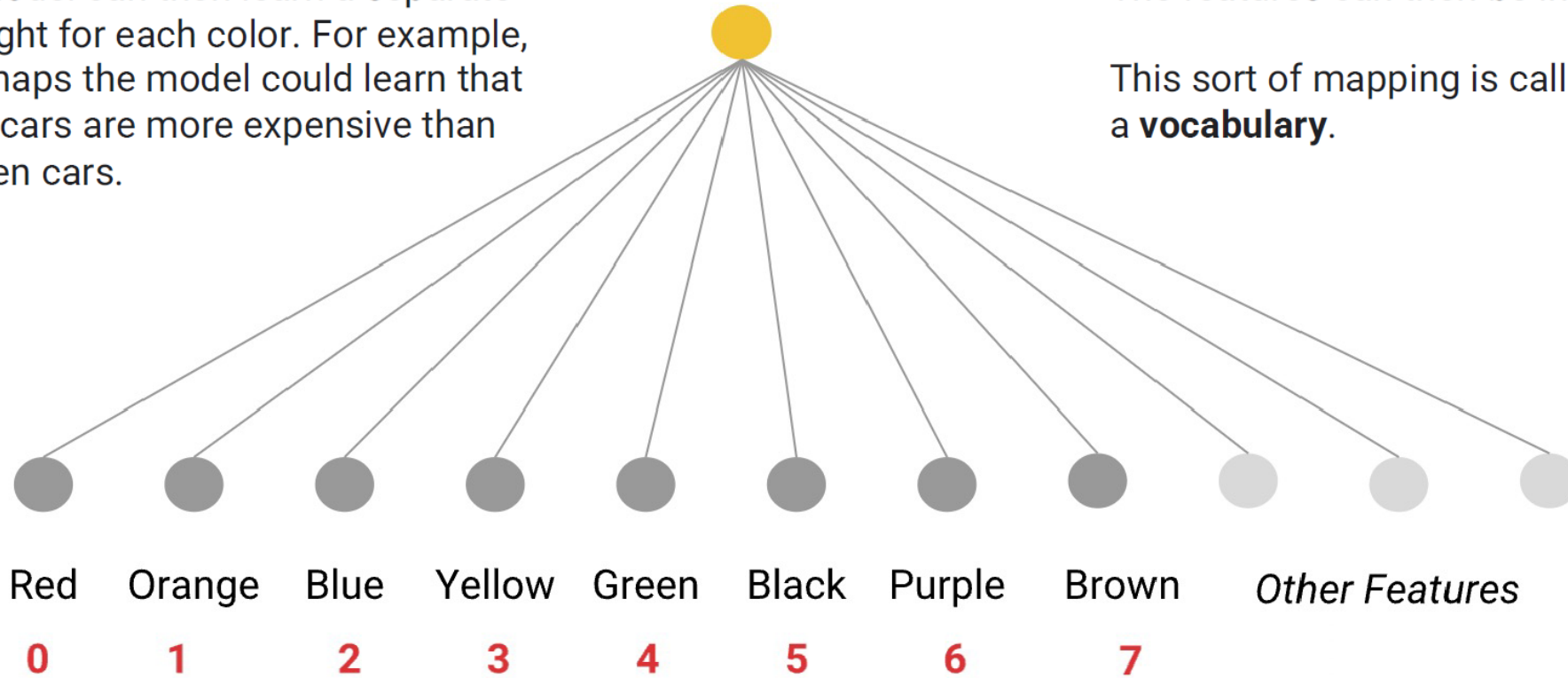


CATEGORICAL DATA

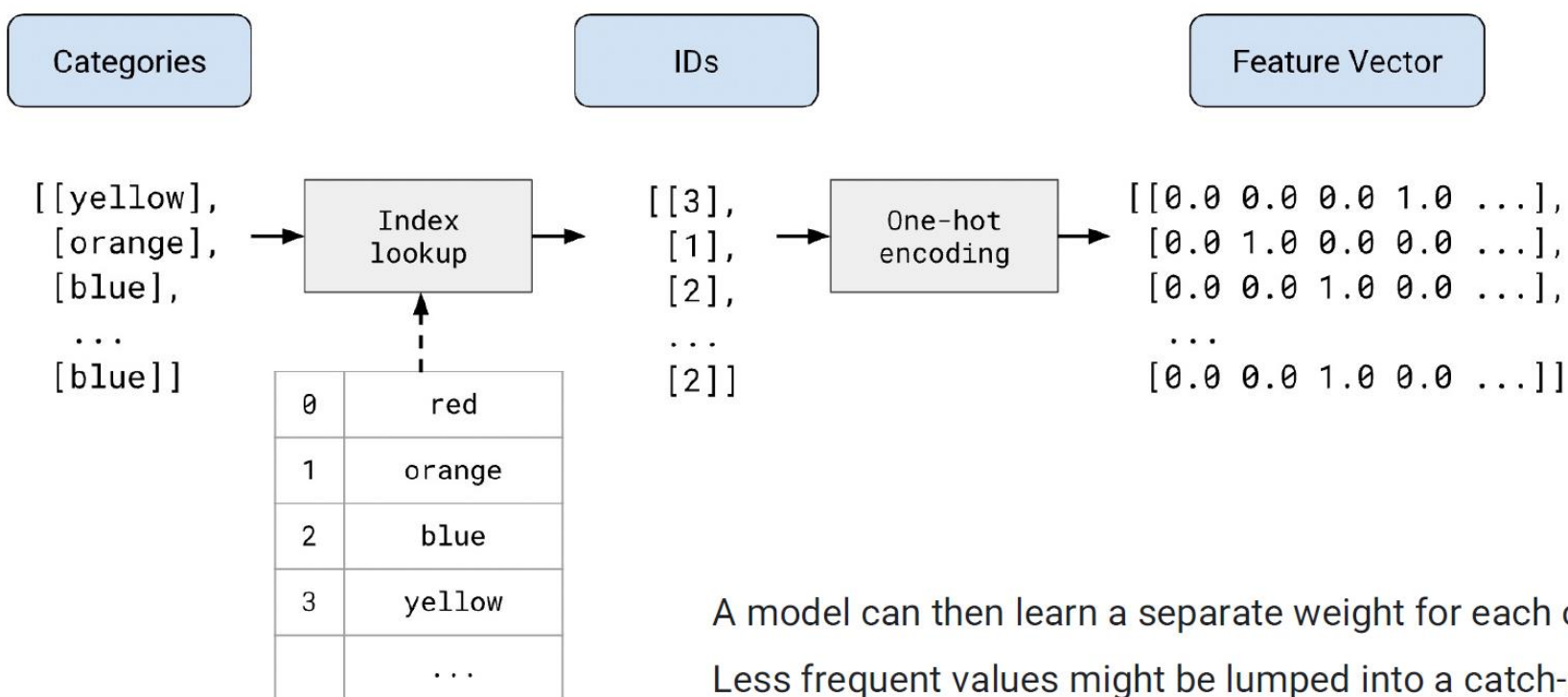
A model can then learn a separate weight for each color. For example, perhaps the model could learn that red cars are more expensive than green cars.

The features can then be indexed.

This sort of mapping is called a **vocabulary**.



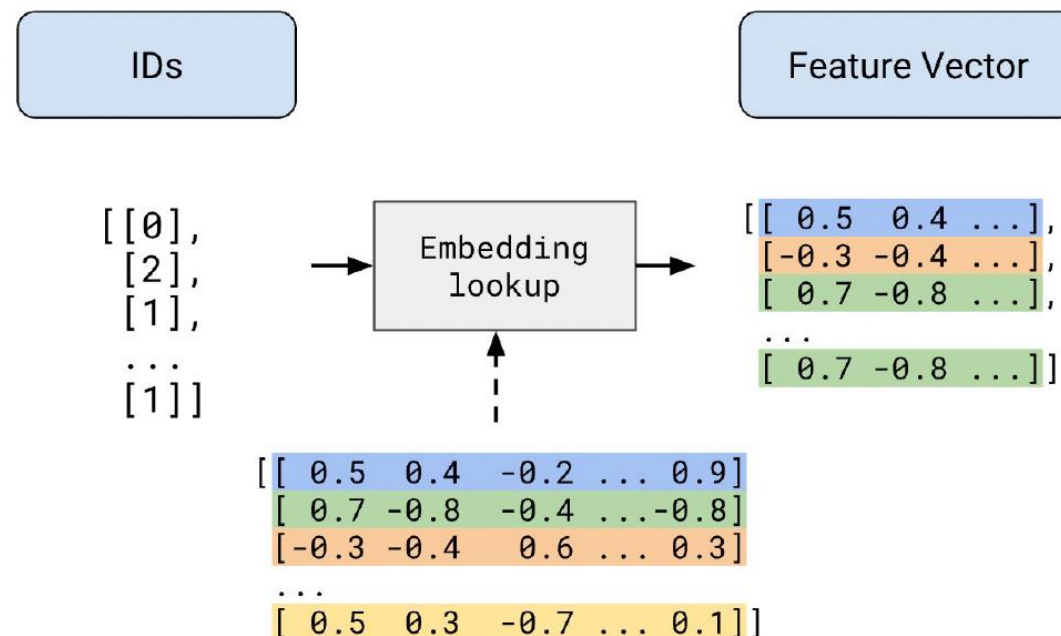
CATEGORICAL DATA: ONE HOT ENCODING



A model can then learn a separate weight for each color
Less frequent values might be lumped into a catch-all category called **Out of Vocab (OOV)**.

CATEGORICAL AND TEXTUAL DATA

- An **embedding** is a relatively low dimensional space into which you can translate high-dimensional vectors
- An embedding captures some of the semantics of the input by placing semantically similar inputs close together in the embedding space
- It is easier to do machine learning on large inputs like sparse vectors representing words.



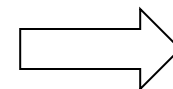
TEXTUAL PREPROCESSING

- Expand Contractions
 - Lower Case
 - Remove Punctuations
 - Remove words containing digits
 - Remove Stopwords
 - Rephrase Text
 - Tokenization
 - Stemming and Lemmatization
 - Remove White spaces
- **Tokenization:** Splitting the sentence into words.
 - **Stop words removal:** Stop words are very commonly used words (a, an, the, etc.) in the documents. These words do not really signify any importance as they do not help in distinguishing two documents.
 - **Stemming:** It is a process of transforming a word to its root form.
 - **Lemmatization:** Unlike stemming, lemmatization reduces the words to a word existing in the language.

FEATURE EXTRACTION

- Encode each input example as a vector in R^n
- There are domain specific techniques according to specific type of data (images, sound, text)

They recently recovered a small piece of a live Elvis concert recording. He was singing gospel songs, including "Peace in the Valley."



0	0	0	They
0	0	1	recently
0	0	2	recovered
0	0	3	a
0	0	4	small
piece	0	5	piece
0	0	6	of
:			
0	1	6	including
0	1	7	QUOTE
peace	1	8	Peace
0	1	9	in
0	1	10	the
0	1	11	Valley
0	1	12	.
0	1	13	QUOTE

0, 1001, 1013, 1134, 1175, 1206
1, 1021, 1055, 1085, 1182, 1252

PRINCIPAL COMPONENT ANALYSIS

- PCA is a dimensionality reduction method that is often used to reduce the dimensionality of large data sets,
- Transform a large set of variables into a smaller one that still contains most of the information in the large set.
- Main steps
 - Standardize the range of continuous initial variables
 - Compute the covariance matrix to identify correlations
 - Compute the eigenvectors and eigenvalues of the covariance matrix to identify the principal components
 - Create a feature vector to decide which principal components to keep
 - Recast the data along the principal components axes

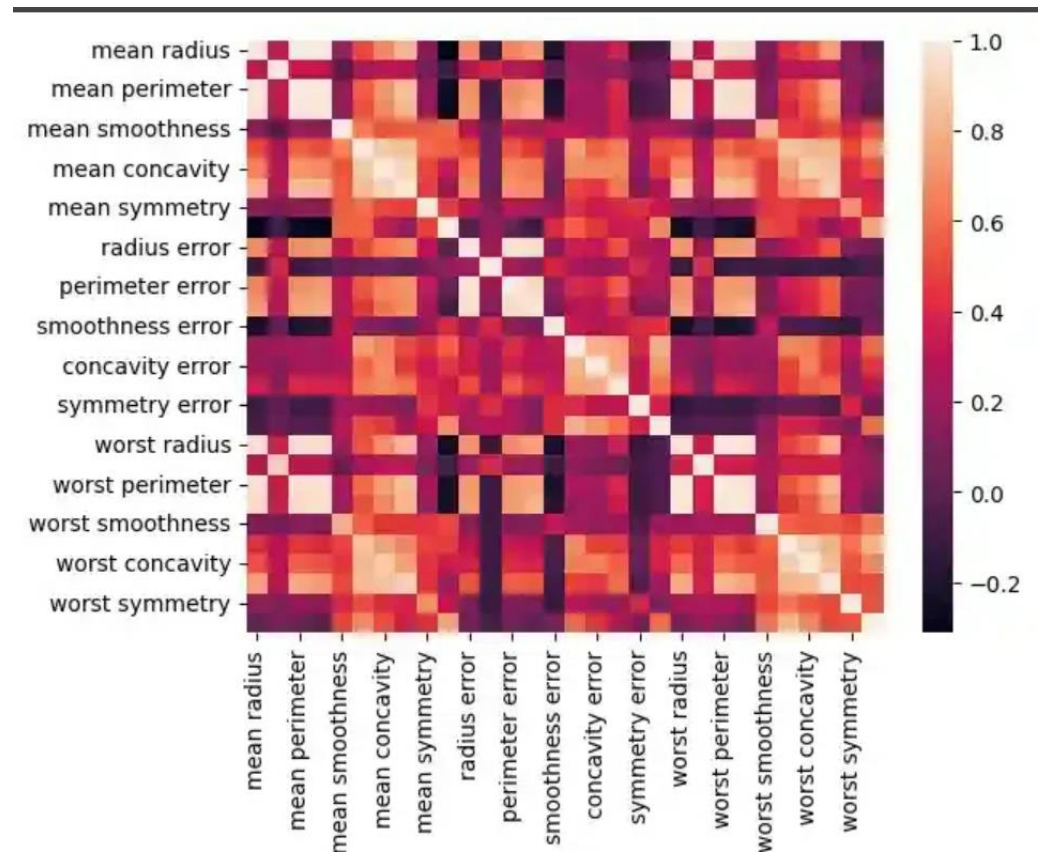
STANDARDIZE THE RANGE OF CONTINUOUS INITIAL VARIABLES

- The aim of this step is to standardize the range of the continuous initial variables so that each one of them contributes equally to the analysis.
- Transforming the data to comparable scales can prevent the sensitivity problem.

$$z = \frac{value - mean}{standard\ deviation}$$

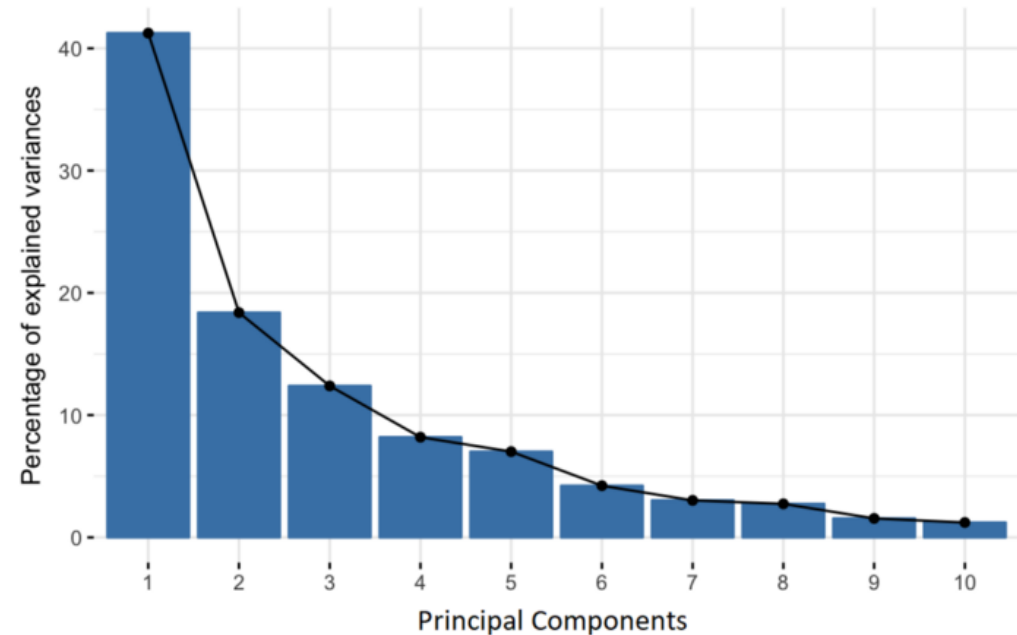
COVARIANCE MATRIX COMPUTATION

- The goal is to understand if there is any relationship between variables
- How the variables of the input data set are varying from the mean with respect to each other
- The covariance matrix is a $p \times p$ symmetric matrix (where p is the number of dimensions) that has as entries the covariances associated with all possible pairs of the initial variables.
- Since the covariance of a variable with itself is its variance ($\text{Cov}(a,a)=\text{Var}(a)$), in the main diagonal there is the variances of each initial variable.
- If the sign of covariance is positive then: the two variables increase or decrease together (correlated)
- If the sign of covariance is negative then: one increases when the other decreases (Inversely correlated)
- The covariance matrix is a table that summarizes the correlations between all the possible pairs of variables



IDENTIFY THE PRINCIPAL COMPONENTS

- Principal components are new variables that are constructed as linear combinations or mixtures of the initial variables.
- These combinations are done in such a way that the new variables (i.e., principal components) are uncorrelated and most of the information within the initial variables is squeezed or compressed into the first components
- In a 10-dimensional data there are 10 principal components, but PCA tries to put maximum possible information in the first component, then maximum remaining information in the second and so on
- principal components are less interpretable and don't have any real meaning since they are constructed as linear combinations of the initial variables.



CONSTRUCT THE PC

- principal components are constructed in such a manner that the first principal component accounts for the largest possible variance in the data set.
- the eigenvectors of the Covariance matrix are the directions of the axes where there is the most variance (most information) and that we call Principal Components.
- Eigenvalues are simply the coefficients attached to eigenvectors, which give the amount of variance carried in each Principal Component.
- Highest eigenvalues is the first principal component
- After having the principal components, to compute the percentage of variance (information) accounted for by each component, we divide the eigenvalue of each component by the sum of eigenvalues.

$$v1 = \begin{bmatrix} 0.6778736 \\ 0.7351785 \end{bmatrix} \quad \lambda_1 = 1.284028$$
$$v2 = \begin{bmatrix} -0.7351785 \\ 0.6778736 \end{bmatrix} \quad \lambda_2 = 0.04908323$$

PC1 and PC2 carry respectively 96 % and 4 % of the variance of the data.

FEATURE VECTOR AND RECAST

- Feature vector is a matrix that has as columns the eigenvectors of the components that we decide to keep.
- If we choose to keep only p eigenvectors (components) out of n , the final data set will have only p dimensions.
- use the feature vector formed using the eigenvectors of the covariance matrix, to reorient the data from the original axes to the ones represented by the principal components
- multiplying the transpose of the original data set by the transpose of the feature vector.

$$FinalDataSet = FeatureVector^T * StandardizedOriginalDataSet^T$$

PCA

plus

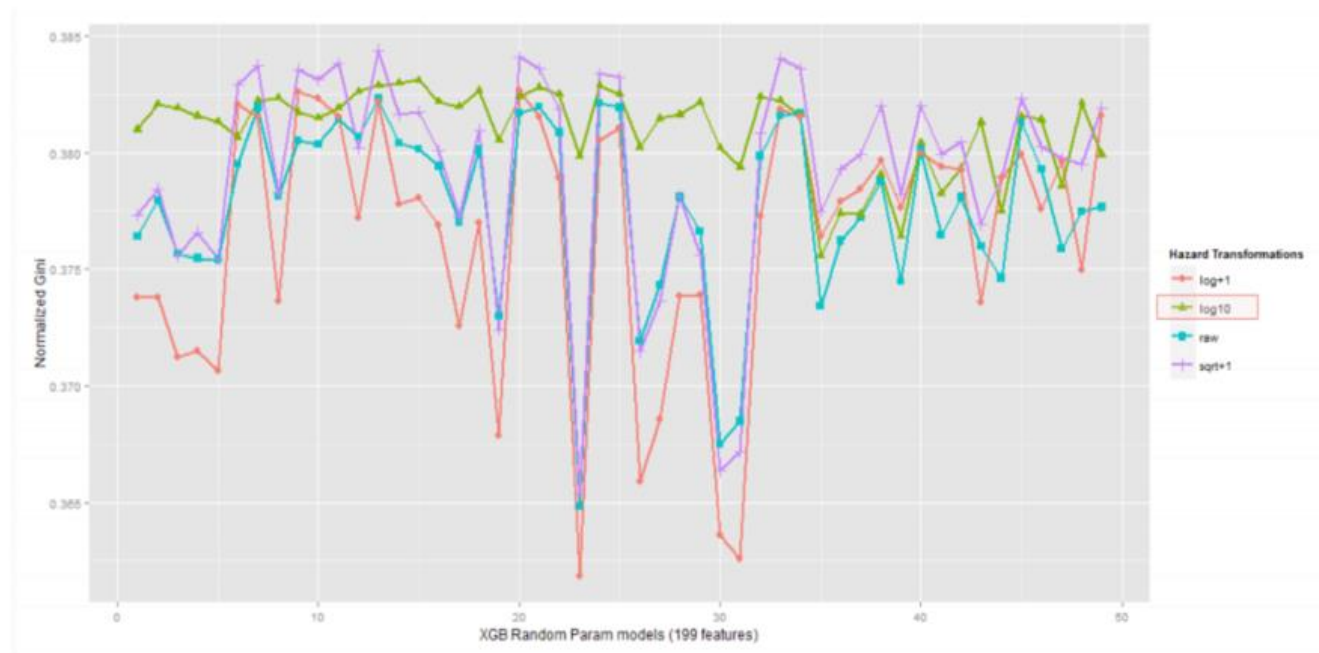
- By reducing the number of variables, PCA simplifies data analysis, improves performance, and makes it easier to visualize data.
- Feature Selection
- Data Visualization: PCA can plot high-dimensional data in two or three dimensions, making it easier to interpret.
- Multicollinearity: PCA can be used to deal with multicollinearity, PCA can help identify the underlying structure in the data and create new, uncorrelated variables that can be used in the regression model.
- Noise Reduction: PCA can be used to reduce the noise in data. removing the principal components with low variance,
- Data Compression
- Outlier Detection

cons

- **Interpretation of Principal Components:**
- **Data Scaling:** Principal Component Analysis is sensitive to the scale of the data.
- **Information Loss:**
- **Non-linear Relationships:**
- **Computational Complexity:** it can be computationally expensive for large datasets.
- **Overfitting:** which is when the model fits the training data too well and performs poorly on new data.

TARGET TRANSFORMATION

- Transform the target variable
 - Cat, dog, fish, apple... $\rightarrow$ 0=others, 1=cat
 - $\text{Log}(x)$, $\text{sqrt}(X)$, ..
- Can improve the model fit
- Different transformation $\rightarrow$ different results



FEATURE SELECTION

- Feature Selection is the method of reducing the input variable to your model by using only relevant data and getting rid of noise in data.
- It is the process of automatically choosing relevant features for your machine learning model based on the type of problem you are trying to solve. We do this by including or excluding important features without changing them. It helps in cutting down the noise in our data and reducing the size of our input data.

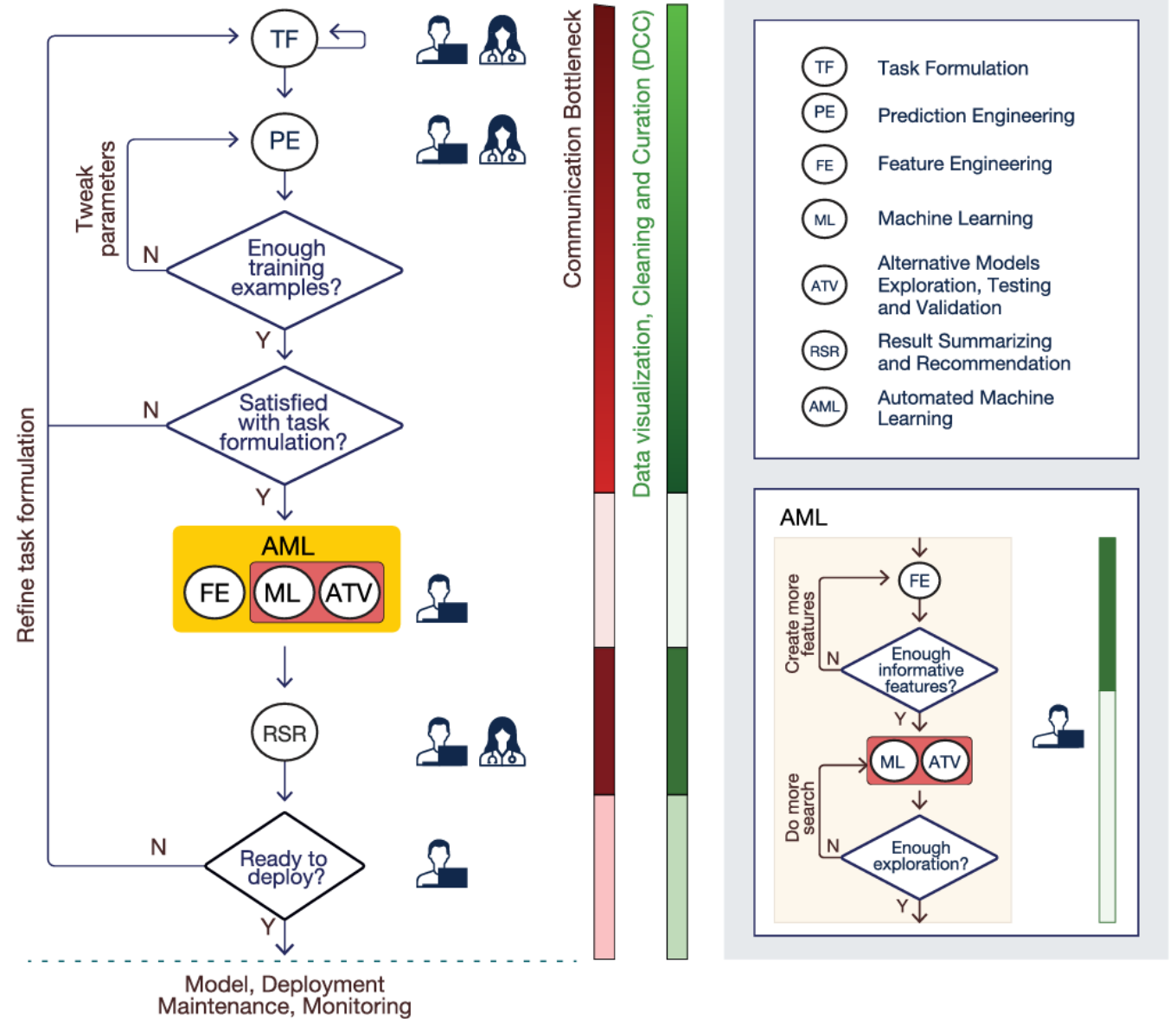


AUTOMATIC FEATURE ENGINEERING



ML VERSUS AUTOML

- ML process and its possible automation



AUTOML

- Different levels of AutoML
- Real-world Example: Consider a wind energy company that owns a number of turbines, each equipped with many sensors. However, numbers generated by the sensors are only meaningful to wind energy experts. A data scientist working on this problem must spend long hours with these experts, to understand the sensor data and what indicates turbine failure so that s/he can create meaningful features. Level 4 tools like MLBazaar aid the data scientist by automating this feature construction step.

	Systems	What is automated?	Access to ML	Efficiency of data scientist
Level 6	???	TF PE AML FE ML ATV RSR		
Level 5	ComposeML + Level 4 systems	PE AML FE ML ATV		
Level 4	Darpa D3M, MLbazaar, RapidMiner	AML FE ML ATV		
Level 3	ATM, Rafiki, Amazon, AutoML, DataRobot, H2O, AUTO-WEKA	AML ML ATV		
Level 2	Scikit-Learn, Keras, Tensorflow, WEKA, ORANGE, Pytorch	ML ATV		
Level 1	Basic implementation of Decision Tree, KMeans, SVM etc.	ML		
Level 0	Programming languages like python, Java, C++			

AUTOMATIC FEATURE SELECTION

- Three different research approaches:
 - Transformation-based
 - Generation-based
 - Learning based

TRANSFORMATION BASES

- Apply different transformations on original features to synthesize new features.
- Katz et al.* identified some basic operators that can transform a feature or combine several to create a new feature.
 - highly informative feature often results from manipulations of elementary ones.”
 - candidate features are fed into the training process, producing a candidate feature that is added to the model based on its empirical performance during the training.
 - Evaluation on 25 datasets and three classification algorithms and shown that they can achieve overall 20% classification-error reduction.
- van den Bosch ** used three different categories of operators (unary, binary, and group-by-them) to transform old features into new ones.
 - a background classifier is used to predict the usefulness of a feature.
 - Feature selection is performed on both original and derived features

• * GILADKATZ, EUICHULRICHARDSHIN, AND DAWN SONG .2016 .EXPLOREKIT: AUTOMATIC FEATURE GENERATION AND SELECTION . ICDM'16 .IEEE,979-984.

• ** SUZANNE VAN DEN BOSCH .2017. AUTOMATIC FEATURE GENERATION AND SELECTION IIN PREDICTIVE ANALYTICS SOLUTIONS. MASTER'S THESIS, FACULTY OF SCIENCE, RADBOUD UNIVERSITY.

GENERATION BASED APPROACH

- Kanter and Veeramachaneni * developed a Deep Feature Synthesis algorithm that can automatically generate features for relational datasets.
 - The algorithm follows relationships in the data to a base field, and then sequentially applies mathematical functions along that path to create the final feature.
 - Intermediate features are optimized by implementing an autotuning process to help it generalize to different datasets.
- Mountantonakis and Tzitzikas ** proposed a generative method based on linked data for discovering, creating, and easily selecting valuable features to describe a set of entities in any ML problem.
 - they first discover datasets and URIs that contain information for a set of entities.
 - they provide users a large number of potential features that can be created for these entities.
 - Finally, they automatically produce a dataset for the features that were selected by the user.
 - They evaluated this approach by performing a fivefold cross validation for estimating the performance of different models for the produced datasets.

• *JAMES MAX KANTER AND KALYAN VEERAMACHANENI. 2015. DEEP FEATURE SYNTHESIS: TOWARDS AUTOMATING DATA SCIENCE ENDEAVORS. DSAA'15

• ** MICHALIS MOUNTANTONAKIS AND YANNIS TZITZIKAS. 2017. HOW LINKED DATA CAN AID MACHINE LEARNING-BASED TASKS. INT. CONF. ON THEORY AND PRACTICE OF DIGITAL LIBRARIES. SPRINGER, 155–168..

LEARNING BASED

- feature generation as a learning task and trained ML models to generate informative features.
- Khurana et al.* uses reinforcement learning (RL) to automate FE.
 - Their strategy is to train an agent on FE examples, enabling the agent to learn an effective strategy to explore available FE choices under a certain budget constraint.
 - This exploration is performed on a directed acyclic graph called “transformation graph,” representing relationships among different transformed versions of the data.
 - Across a variety of datasets, they have shown that models produced by their system reduced median error rate by 25%.
- Kaul et al.** presented a regression-based feature learning algorithm called “AutoLearn.”
 - They apply regression on each feature to predict the values of other feature and use this regression forecast to supplement additional information in each record.
 - The authors have shown that, compared to original feature space, features learned through their model can improve prediction accuracy by 13.28% and 5.87% with respect to other top performing models.

• * UDAYAN KHURANA, HORSTSA MULOWITZ, AND DEEP AKTURAGA. 2017. FEATURE ENGINEERING FOR PREDICTIVE MODELING USING REINFORCEMENT LEARNING. ARXIV:1709.07150.

• ** AMBIKA KAUL, SAKET MAHESHWARY, AND VIKRAMPUDI. 2017. AUTOLEARN- AUTOMATED FEATURE GENERATION AND SELECTION. ICDM'17 .IEEE, 217–226.



TOOLS



TOOLS

- ExploreKit is an automated feature generation kit that is essentially an implementation of the transformation-based approach introduced by Katz et al.
- FeatureTools is an automated feature engineering framework based on the Deep Feature Synthesis method developed by Kanter and Veeramachaneni . FeatureTools can effectively perform feature engineering for relational and temporal data.They also provide an R implementation named featuretoolsR
- Featexp is a python package for feature exploration that shows graphs and plots to help users better understand the feature space. Featexp bins a feature into equal population bins and shows the mean value of the dependent variable (target) in each bin, the trend of which can help users understand the relationship between the target and the feature. Users can compare feature trends to identify noisy features by looking at the number of trend direction changes and the correlation between train and test trend.These graphs/plots also help user to do feature debugging and leakage detection
- MLFeatureSelection has broken feature selection into three parts. First, it selects a sequence from raw data that will potentially contain the best feature combination. Second, it creates feature groups from the raw data and iteratively update them and remove less important features. Eventually, this process yields a set of important features.Third, a coherent threshold is picked and all the features having higher coherence value are selected as best possible features.
- Feature Engineering & Feature Selection provides a guideline for feature engineering and feature selection techniques including why to use them, when to use them and when to use which algorithm.
- FeatureHub provided a collaborative framework for feature engineering and instantiate this idea into a platform.This platform administers collaboration among data scientists where users can register features in the database as well as reuse features written by other users. FeatureHub automatically scores the features so users can get real-time feedback for their features.
- Feast introduces a centralized feature store that can work as a foundation of features used by different projects. Feast allows users to use historical feature data to produce sets of features for training ML models, and makes feature data available to models in production by serving API and ensuring consistency between training and serving is preserved